

RESCUE-NET

MASTER ARCHITECTURE & DEVELOPMENT HANDBOOK

VOLUME 13

PRODUCT ROADMAP & NATIONAL SCALING

Versi 1.0 | 1 Agustus 2026
Internal Project Reference

Kendali Dokumen

Atribut	Nilai
Judul	Rescue-Net Volume 13 - Product Roadmap & National Scaling
Versi	1.0
Tanggal	1 Agustus 2026
Status	Roadmap produk dan scaling baseline
Horizon	2026-2030
Fokus	Prioritas produk, capability maturity, rollout wilayah, governance, KPI, risiko, pembiayaan, dan national-scale architecture

Prinsip roadmap

Roadmap tidak boleh mengejar jumlah fitur. Prioritas selalu dimulai dari kestabilan database, kualitas data, konsolidasi, offline reliability, War Room, lalu scaling.

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1. Tujuan Roadmap

Roadmap ini memberi urutan implementasi agar Rescue-Net berkembang dari sistem yang sedang distabilkan menjadi platform koordinasi bencana lintas wilayah tanpa mengorbankan kualitas data dan auditabilitas.

- Menentukan urutan prioritas yang realistis.
- Membedakan capability wajib, enhancement, dan future option.
- Menghindari big-bang transformation.
- Menetapkan milestone teknis, operasional, dan kelembagaan.
- Menyiapkan jalur menuju deployment multi-kabupaten, multi-provinsi, dan nasional.

2. Visi Produk

Rescue-Net diarahkan menjadi platform koordinasi dan decision support yang mengubah laporan mentah menjadi Operational Truth yang dapat ditelusuri. Nilai utamanya bukan banyaknya laporan, melainkan kualitas keputusan, kecepatan respons, keterlacakan bantuan, dan kemampuan bekerja saat internet terbatas.

North Star

Satu operational picture yang dapat dipercaya, terus diperbarui, dan dapat dijelaskan sumbernya.

3. Baseline dan Kondisi Awal

Area	Baseline
Deployment	Synology NAS, Docker, nginx, Tailscale Funnel
API	rescue-net-api pada port 8092
Database	postgres-main / rescuenet_db
Web	/rescue-net/
PWA	/volume1/web/rescue-net-app
Trusted Verifier	Tabel dan endpoint dasar tersedia
Consolidation	Scenario minimum/optimal/maximum tersedia sebagian
War Room	Scenario buttons dan summary endpoint tersedia
Known risk	Database connectivity dan reproducibility pernah bermasalah

3.1 Gap Utama

- Database configuration belum cukup reproducible.
- Observability dan restore drill belum matang.
- Wilayah resmi belum terintegrasi penuh.
- Offline sync dan conflict handling perlu diperkuat.
- War Room drill-down dan trace perlu diselesaikan.
- Packaging aplikasi belum siap rilis publik penuh.

4. Prinsip Prioritas

Urutan	Prinsip
1	Stability before scale
2	Data quality before analytics
3	Consolidation before distribution
4	Offline reliability before native expansion
5	Traceability before automation
6	Pilot before national rollout
7	Capability before menu

5. Product Capability Map

Capability	Target akhir
Public Reporting	Multi-channel, offline-capable, traceable
Verification	Identity, location, organization, evidence, need terpisah
Trusted Verifier	Trust network dengan scope dan revoke
Consolidation	Duplicate/overlap/conflict engine
Operational Truth	Versioned, confidence-based, auditable
War Room	Drill-down dan decision workspace
Resources	Stock, transport, volunteer, medical, logistics
Distribution	Plan, dispatch, delivery, confirmation
Mobile/Offline	Reliable queue, retry, conflict handling
GIS	Wilayah resmi, map, spatial analysis
AI	Assistance, bukan authority
Analytics	Trend, gap, performance, forecast

6. Maturity Model

Level	Deskripsi
1 - Fragmented	WA, spreadsheet, data terpisah
2 - Digital Reporting	Laporan masuk ke platform
3 - Verified	Identity, location, evidence, organization
4 - Consolidated	Duplicate dan overlap dikendalikan
5 - Decision Support	War Room dan planning
6 - Integrated	Lintas organisasi dan wilayah
7 - National	Federated national operational picture

Target jangka dekat Rescue-Net adalah Level 4-5 pada pilot terbatas sebelum mengejar Level 6-7.

7. Roadmap 2026 - Stabilization and Core Completion

Quarter	Fokus	Deliverable
Q3 2026	Stabilitas runtime	DB connection, backup, restore drill, readiness, logging
Q3 2026	Data baseline	Schema inventory, ERD aktual, migration audit
Q4 2026	Wilayah	Master wilayah resmi, cascading UI, offline cache
Q4 2026	Offline	Queue, idempotency, retry, conflict UI
Q4 2026	Consolidation	Drill-down, trace source, command correction

7.1 Exit Criteria 2026

- Database dapat direcreate tanpa kehilangan konfigurasi.
- Backup berhasil direstore.
- Public reporting dan offline sync lulus UAT.
- Overlap same beneficiary default MAX.
- War Room menampilkan trace dari source ke consolidated need.

8. Roadmap 2027 - Pilot Operations

Fokus	Deliverable
Pilot wilayah	1-3 kabupaten dengan use case nyata
Operational SOP	Operator, verifier, posko, command center
Resource module	Stock, vehicle, volunteer, medical
Distribution	Assignment, route, delivery confirmation
Mobile release	Android artifact signed dan diuji
Analytics	Operational KPI dan after-action report

8.1 Pilot Criteria

- Ada sponsor/owner operasional lokal.
- Ada data wilayah dan organisasi.
- Ada scenario exercise atau event nyata terkontrol.
- Ada operator yang dilatih.
- Ada after-action review dan issue backlog.

9. Roadmap 2028 - Multi-Region Integration

Fokus	Deliverable
Multi-region	Beberapa kabupaten/provinsi
Federation	Regional operational nodes dan national rollup
Integration	Sumber data pemerintah/NGO yang disetujui
High availability	Separate DB/storage dan standby strategy
Data governance	Owner, steward, quality rules
Security	Formal access review dan incident response

10. Roadmap 2029 - Decision Intelligence

Fokus	Deliverable
Forecast	Need and resource forecast sebagai rekomendasi
Simulation	Scenario planning untuk resource shortage
Knowledge	Lesson learned dan pattern library
AI copilot	Operator assistant dengan provenance
GIS intelligence	Risk layer, access route, coverage gap

11. Roadmap 2030 - National Platform Readiness

Fokus	Deliverable
National interoperability	Standar API dan data exchange
Federated governance	Pusat, provinsi, kabupaten, organisasi
National DR	Multi-region recovery
Certification	Security, privacy, operational assurance
Sustainability	Operating funding dan support model

12. Pilot dan Rollout Wilayah

30. Lab/staging validation.
31. Tabletop exercise.
32. Single-kabupaten pilot.
33. Multi-kabupaten pilot.
34. Provincial rollup.
35. Inter-provincial federation.
36. National operational integration.

12.1 Rollout Gate

Gate	Syarat
Technical	Health, backup, security, performance
Data	Wilayah, organization, posko, verifier
Operational	SOP dan trained users
Governance	Owner dan escalation path
Legal/privacy	Data handling dan sharing agreement

13. National Scaling Architecture

Regional Nodes

- |-- local reporting
- |-- local verification
- |-- local operations

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National Coordination Layer

- |-- federated rollup
- |-- cross-region resource view
- |-- standards and governance
- |-- national situational awareness

13.1 Scaling Principles

- Regional operations tetap berjalan ketika koneksi pusat terganggu.
- National layer tidak menghapus local provenance.
- Data sharing mengikuti minimum necessary principle.
- Regional correction dan national correction memiliki authority yang jelas.

14. Operating Model Nasional

Level	Peran
National	Standards, national rollup, cross-region coordination
Provincial	Regional coordination dan escalation
District/City	Operational command dan data stewardship
Village/Posko	Field capture, verification, distribution evidence
Organization	Resource, verifier, operational contribution

15. Governance dan Ownership

Domain	Owner
Product	Product steering group
Architecture	Lead architect/technical authority
Data	Data governance board dan domain stewards
Security	Security owner
Operations	Command/operations owner
Platform	Engineering and DevOps
AI	AI governance owner

15.1 Decision Rights

- Product owner menetapkan prioritas bisnis.
- Architecture authority mengendalikan perubahan fundamental.
- Command owner menetapkan operational corrections.
- Data steward menetapkan quality rules.
- Security owner dapat memblokir release berisiko tinggi.

16. KPI dan OKR

Area	KPI
Reliability	Availability, error rate, restore success
Data quality	Duplicate rate, unresolved conflict age
Verification	Time to verify dan revoke response
Offline	Sync success dan oldest queue age
War Room	Time from report to consolidated need
Distribution	Plan-to-delivery time dan confirmation rate
Adoption	Active operators, posko, organizations
Trust	Correction trace completeness

16.1 North Star Metrics

- Median time dari raw report ke operational decision.
- Persentase kebutuhan yang memiliki source trace lengkap.
- Persentase distribusi dengan delivery evidence.
- Persentase sync offline berhasil tanpa duplikasi.

17. Risk Register

Risiko	Dampak	Mitigasi
Database/config drift	API down	Reproducible config dan restore drill
Over-automation	Keputusan salah	Human-in-the-loop
Duplicate beneficiary	Over-distribution	MAX default dan operator review
Weak offline sync	Data hilang/duplikat	Idempotency dan conflict UI
Unclear governance	Keputusan tidak sah	Decision rights
Premature national rollout	Failure at scale	Pilot gates
Privacy breach	Trust/legal risk	Data minimization dan access control
Dependency on one host	Outage risk	HA roadmap

18. Technical Debt Roadmap

Debt	Priority	Action
Runtime not fully reproducible	P0	Document and automate
Schema not fully inventoried	P0	Live ERD and migration ledger
Limited observability	P0	Logs, metrics, alerts
Offline conflict gaps	P1	State machine and UI
War Room read model	P1	Optimize and trace
Packaging uncertainty	P2	Release pipeline
Microservice temptation	Avoid	Stay modular until proven

19. Data Governance Nasional

- Satu master wilayah resmi.
- Source version dan update date wajib.
- Data owner dan steward per domain.
- Quality rules dan exception workflow.
- Retention dan archive policy.
- Data sharing agreement lintas institusi.

20. Security dan Compliance Roadmap

Tahap	Fokus
2026	Secrets, least privilege, backup, audit
2027	Access review, incident response, privacy policy
2028	Penetration testing dan regional security model
2029	Formal security governance dan supplier assurance
2030	National-scale compliance readiness

21. AI Roadmap

Tahap	Use Case
Foundation	OCR, classification, summarization
Assistive	Duplicate candidate dan evidence signal
Predictive	Forecast need/resource
Decision support	Scenario recommendation dengan provenance

Guardrail

AI tidak boleh menetapkan Operational Truth, menolak laporan, atau menentukan distribusi tanpa human authority.

22. GIS Roadmap

- Master wilayah resmi.
- GPS/map/admin-area verification.
- Spatial duplicate detection.
- War Room drill-down dan heatmap.
- Access route dan resource coverage.
- Risk and forecast layers.

23. Mobile dan Offline Roadmap

Tahap	Deliverable
Foundation	PWA offline, local queue
Reliability	Idempotency, retry, conflict
Release	Android signed build
Expansion	iOS build flow
Field maturity	Offline map, assignment, delivery evidence

24. Frappe Hybrid Migration

Pindah lebih dulu	Tetap custom
Master user/role	Consolidation engine
Organization/posko admin	Operational Truth
Workflow approval	Offline sync
Back-office reporting	War Room real-time

24.1 Gate Migrasi

- Database existing stabil.
- Backup dan restore lulus.
- Integration contract jelas.
- Pilot noncritical di staging.
- Tidak ada direct cross-database coupling tanpa governance.

25. Partnership Strategy

Partner	Kontribusi
Government	Authority, data, operations
NGO/Community	Field network dan validation
Universities	Research, training, UAT
Technology partners	Infrastructure, security, AI/GIS
Donors/CSR	Funding dan equipment

26. Funding dan Sustainability

- Core platform funding.
- Implementation and training funding.
- Support and maintenance model.
- Disaster surge budget.
- Independent audit and security budget.

Sustainability tidak boleh bergantung hanya pada proyek awal. Operating cost, support, training, data maintenance, dan DR harus memiliki model pembiayaan.

27. UAT dan Certification

37. Functional UAT.
38. Offline and reconnect UAT.
39. Duplicate/overlap scenario UAT.
40. Security testing.
41. Backup/restore drill.
42. Operational exercise.
43. Pilot acceptance.
44. Scale readiness review.

28. Immediate Next Actions

45. Selesaikan audit database dan runtime.
46. Buat roadmap backlog 90 hari.
47. Tetapkan owner per capability.
48. Selesaikan wilayah master dan offline sync.
49. Lengkapi War Room trace dan conflict queue.
50. Pilih satu wilayah pilot.
51. Susun UAT scenario dan operator training.
52. Review roadmap setiap quarter.

Prioritas mutlak

Tidak ada national rollout sebelum sistem dapat dibackup, direstore, dan menjalankan offline sync tanpa duplikasi.

29. Acceptance Criteria

- Roadmap memiliki owner dan milestone.
- P0 stability selesai sebelum scale.
- Pilot memiliki exit criteria.
- KPI dapat diukur dari sistem.
- Risk register ditinjau berkala.
- Frappe migration mengikuti gate.
- AI dan GIS mengikuti governance.
- National scaling mempertahankan regional resilience.

Lampiran A - Quarterly Review Template

Area	Pertanyaan
Outcome	Apa yang berubah untuk operasi?
Reliability	Apakah uptime, restore, dan sync membaik?
Data	Apakah conflict dan duplicate menurun?
Adoption	Siapa yang aktif menggunakan?
Risk	Risiko baru apa yang muncul?
Next quarter	Tiga prioritas tertinggi?

Lampiran B - Architecture Decision Records

ID	Keputusan
ADR-RM-001	Stability before scale
ADR-RM-002	Pilot-gated national rollout
ADR-RM-003	Regional resilience with national rollup
ADR-RM-004	Capability-based prioritization
ADR-RM-005	Hybrid Frappe migration